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How the engine *reads* the market.

The complete internal reference for the RigaCap signal pipeline — the data, the indicators, the entry ensemble, the regime engine, and the discipline that calibrates one system into two tiers.

MARKET REGIMES	PRODUCT TIERS	SYMBOLS	WALK-FORWARD
7	2	~4,000	21y
Read daily	One engine, two dials	Scanned daily	Continuous, no cherry-picking

CONTENTS

What this document *covers.*

A technical blueprint for the RigaCap signal engine: how raw market data is acquired, how indicators are computed, how the entry ensemble decides, how the regime engine shapes behavior, and how the shared base engine — plus the Maximizer breakout book — becomes the Preserver and Maximizer tiers subscribers actually buy.

PORTFOLIO & OPERATIONS

One sentence on posture. Every performance figure in this document comes from a single continuous walk-forward path — what one investor would actually have experienced across a full cycle, not a flattering window. Production parameters are locked. Where the walk-forward harness has research-mode capabilities beyond the live rules, those are flagged explicitly. This is the internal blueprint: it documents the real machinery, including the mechanics that never appear in public materials.

01 · FOUNDATION

Two tiers, *one engine.*

RigaCap is a single momentum engine with a risk dial. Everything downstream — data, indicators, entry ensemble, regime read — is shared. What differs between the two tiers is how aggressively the engine raises cash when the market capitulates.

Publicly we call them **Preserver** (protect) and **Maximizer** (grow). They share one base engine — the same universe, the same ranking, the same trailing stop, the same regime classifier — and they diverge in two deliberate ways. First, the capitulation overlay (Section 08) fires harder for Preserver and lighter for Maximizer. Second, Maximizer layers a separate gated breakout book (Section 09) on top of the shared base; Preserver does not. Two levers on one shared foundation — which is why parity discipline (Section 13) matters so much: a defect in the shared engine touches both products.

Preserver

PROTECT · FLAGSHIP BASE

13.0%	1.28	−12.9%
5YR CAGR	SHARPE	MAX DD

Momentum built around the drawdown. Raises cash aggressively in capitulation; gives up a little in smooth melt-ups — the insurance premium.

Maximizer

GROW · AGGRESSIVE ADD-ON

31.4%	1.51	−14.9%
5YR CAGR	SHARPE	MAX DD

Offense with a seatbelt. The Preserver base plus a separate gated breakout book (Section 09), sized by a volatility target — higher return with a tamed momentum-crash tail.

5-year walk-forward figures (vintage 2026-07-31). SPY over the same window: 14.2% CAGR, −25.4% max drawdown. Preserver and Maximizer share one base engine; they differ in the capitulation-overlay intensity (Section 08) and in the gated breakout book that only Maximizer runs (Section 09).

The design principle. Marketing's job is filtration, not conversion — the same engine is an F for the wrong investor and an A for the right one. Preserver is for capital-preservers who would panic-sell a 50% drawdown; Maximizer is for growth-seekers who can stomach a rough year. We tell each their tier's cost up front.

The data *pipeline.*

The pipeline begins with a universe of roughly four thousand US equities. Every trading afternoon the system fetches, validates, and enriches the price record before it ever reaches the signal engine.

Stock universe.

PARAMETER	VALUE	SOURCE
Total universe	`~4,000 symbols`	`NASDAQ + NYSE`
Minimum price	`$15`	`config.MIN_PRICE`
Minimum volume	`500,000 shares/day`	`config.MIN_VOLUME`
Lookback	`7 years OHLCV`	`Parquet on S3 (per-symbol PITFWU store); pickle legacy`
Primary source	`Alpaca SIP`	`Consolidated tape; stored as-traded, adjusted at read`
Fallback	`yfinance`	`Indexes (^VIX, ^GSPC), failover`

Universe hygiene.

All exchange-traded funds are excluded from the signal universe. The exclusion is categorical — leveraged products, sector ETFs, broad-index ETFs, crypto-linked products, and thematic baskets are all out. Ordinary operating companies, including miners, are eligible. The exclusion is enforced at the symbol-list level and reasserted at scan time so a recategorised ticker cannot leak into the universe between rebuilds.

Fetch architecture.

PARAMETER	VALUE	PURPOSE
Batch size	`100 symbols (Alpaca) / 10 (yfinance)`	`Provider-appropriate batching`
Max retries	`2 per batch`	`Transient failure handling`
Batch delay	`1.5 s`	`Rate-limit compliance`
Retry delay	`3.0 s (2x batch delay)`	`Backoff on failure`
Required symbols	`SPY, ^VIX`	`Always fetched (regime detection)`

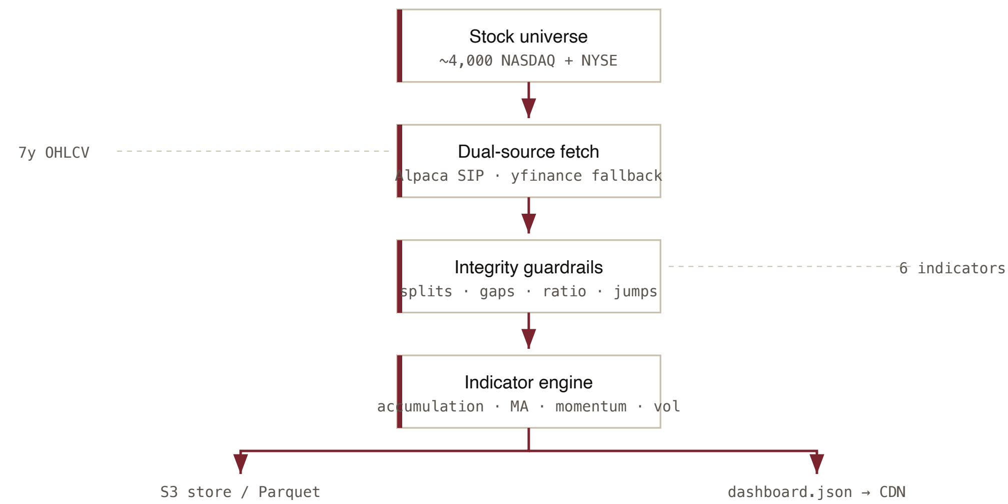
After the initial multi-year fetch, daily updates run in **incremental mode** — only prices since the last cached date are downloaded. Updates that took minutes on a cold load take seconds incrementally. The full historical record is persisted to S3 as **Parquet**: a point-in-time, split-adjusted per-symbol store is the live read path — scoped, partial reads keep Lambda memory low — with `all_data.parquet` as the base and the fallback for missing or short-history symbols, queryable ad-hoc via DuckDB. A legacy consolidated gzipped pickle is retained only as a cold-start fallback and is being retired.

Data-integrity guardrails.

Split-adjustment and corruption detection are built into every layer of the pipeline. End-to-end split-adjustment eliminates a class of historical-data artifacts that would otherwise distort returns or trigger spurious exits.

- **Split-adjusted by default.** The Alpaca `StockBarsRequest` uses `Adjustment.SPLIT` on every call, so a 10-for-1 or 20-for-1 split cannot create an artifact –90% drop in the historical series.
- **Abrupt-jump rejection.** Any single-day close-to-close change greater than 65% (log-return > 0.5) marks the symbol as corrupted. Real market moves rarely exceed this; unadjusted corporate actions always do.
- **Ratio sanity.** Symbols where the current close is more than 50x or less than 0.02x the long-term accumulation reference are quarantined as clearly corrupted — typically ticker reuse or a missed adjustment.
- **Long-gap rejection.** Any symbol with a gap longer than ten trading days between consecutive bars is flagged as ticker-reuse corruption (old entity delisted, ticker recycled to a new entity).
- **Nightly Layer 2 hygiene.** Alpaca's corporate-actions API is polled every weekday at 6:30 PM ET. Splits trigger automatic force-refetch with split adjustment. Ticker reuses trigger auto-quarantine. An admin digest reports outcomes.
- **Asset-ID tracking.** Every symbol's Alpaca UUID is stored in `symbol_metadata`. When an exchange recycles a ticker, the UUID changes and the system detects the swap before the next signal fires.

Data pipeline flow



From raw bars to a single source-of-truth dashboard.

The point-in-time *price store.*

This is the quiet piece of engineering that makes every number in this document trustworthy. Most backtests are contaminated before they begin — not by bad logic, but by bad data. Our price store is built to remove that contamination at the source.

The problem it solves.

Nearly every market-data vendor serves an *adjusted* close — a price silently rewritten backwards through history every time a stock splits or pays a dividend. Convenient for a chart; poison for a walk-forward. It means a simulation run today “sees” prices no investor could actually have seen at the time — a subtle look-ahead that flatters results and quietly inflates the very edge you are trying to measure.

What we do instead.

We store the **raw, as-traded bars** — the actual prices that printed on the tape that day — one file per symbol (internally, the **PITFWU** store), alongside a **corporate-actions calendar** of splits and dividends with their effective dates. Adjustment is applied **at read time**, point-in-time: when the system asks for a symbol’s history as of a given date, it reconstructs exactly what the tape looked like *on that date* — no future split has leaked backwards, no next-year dividend has touched last year’s price. The data a strategy trades on in research is, bar for bar, the data a subscriber’s decision would have faced in real life.

> **Why it’s the real magic.** Research and production read the *same* point-in-time store through the *same* read veneer, so the walk-forward result and the live book reconcile penny-to-penny. The published numbers aren’t a lab curiosity that degrades in the wild — they are produced by the identical data path a live

> subscriber runs. That parity is only possible because the data itself carries no hindsight.

Built to be cheap and safe.

The store is per-symbol on purpose. A daily update appends only today's bars to the symbols that traded — no rewrite of a multi-hundred-megabyte monolith, which is what used to risk running the pipeline out of memory. Reads are **scoped**: the daily scan pulls only the few hundred symbols it actually needs, never the full universe history, so memory stays flat no matter how many years accumulate. A consolidated base file (`all_data.parquet`) backs up any symbol missing or too new to have its own history, and a legacy pickle cache is retained only for cold-starts while it is retired. It is, deliberately, the **single source of truth** — one store, raw and honest, adjusted consistently at read, feeding research and production alike.

The indicator *engine.*

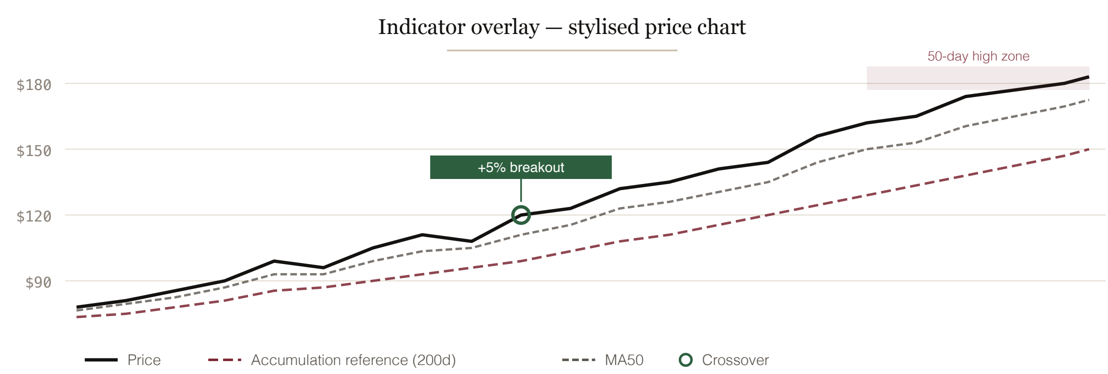
A short list of indicators — computed once per scan, cached on demand — feeds both the timing layer and the momentum-quality ranking.

INDICATOR	FORM	PERIOD	ROLE
Accumulation reference	`Σ(price × volume) / Σ(volume)`	200d	Volume-weighted timing baseline
Moving averages	Simple	20 / 50 / 200d	Trend confirmation, quality filter
Short momentum	`(p / p[t-5] - 1)`	5d	Short-term acceleration · weight 0.3
Long momentum	`(p / p[t-60] - 1)`	60d	Sustained trend strength · weight 0.2
Volatility	`σ(returns,20) × √252`	20d	Risk penalty in composite · weight 0.15
Distance from high	`p / max(p,50d) - 1`	50d	Breakout proximity

> **Lazy computation.** Indicators are computed on demand and cached in the per-scan data structure. The first scan pays for the indicator pass; subsequent

operations within the same scan reuse cached values.

The accumulation reference is RigaCap's proprietary timing baseline — a long-term volume-weighted price that filters out noise from short-term spikes. The breakout filter fires when price clears that baseline by a small, conviction-weighted margin and the rest of the indicator stack confirms.



A timing trigger plus trend confirmation — both have to fire.

The entry ensemble — *three* factors.

A candidate must satisfy three independent factors before the system generates an entry. Multi-confirmation reduces false signals and keeps the bar high. Both tiers use the same entry gauntlet — the tiers diverge only on the exit/cash-raise side.

Factor 1 — timing.

A proprietary breakout detector fires when price clears the long-term accumulation reference by a conviction-weighted margin. The threshold is a fixed lever in production (the 5% level), tuned during walk-forward research and locked at deployment. It catches early breakouts before momentum-only systems acknowledge them.

Factor 2 — momentum quality.

Candidates that pass timing are ranked by a composite momentum score combining short-term acceleration (5-day), longer-term trend strength (60-day), and a volatility penalty. The production formula, exactly as it runs live:

$$\text{score} = 0.3 \times r_{5d} + 0.2 \times r_{60d} - 0.15 \times \sigma_{20d}$$

where r_{5d} and r_{60d} are trailing returns in percent and σ_{20d} is 20-day annualised volatility in percent. Only top-ranked names are eligible. Note the weighting: the **short horizon carries the higher weight** — the composite rewards fresh acceleration within an established 60-day trend, while the volatility penalty disqualifies candidates whose "momentum" is really just whipsaw. Candidates must additionally trade above their 20-day and 50-day moving averages (the trend-quality gate).

Factor 3 — confirmation.

FILTER	CONDITION	PURPOSE
Breakout proximity	Within 3% of 50-day high	High-conviction entries only
Volume confirmation	Day's volume ≥ 1.3x recent average	Real participation
Trend	Price > MA20 and MA50	Established uptrend
Liquidity	≥ 500,000 shares/day	Tradable size
Floor	Price ≥ $15	Filters penny stocks

Anti-squeeze filters.

A fourth filter family rejects parabolic candidates that have already run before our other factors fire — a guard against entering at squeeze tops. The filter was added after a review of the strategy's worst historical entries.

FILTER	LEVER	PURPOSE
Recent-return cap	`max_recent_return_pct`	Reject candidates already up beyond a 30-day threshold
Velocity cap	`price_velocity_cap_pct`	Reject candidates with an extreme 5-day move
Overbought guard	`rsi_oversold_filter`	Standard RSI ceiling on the entry candidate

Production parameters.

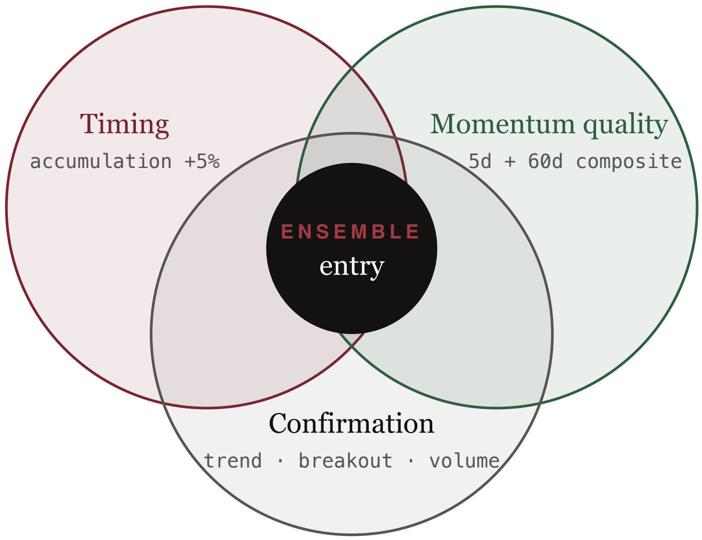
Base book (both tiers). Production trades a locked parameter set: up to twenty positions sized at roughly 4.5% each with inverse-volatility weighting, a 30% trailing stop, and the timing-and-momentum ensemble described above, drawn from a universe of the most-liquid symbols by 60-day volume. The walk-forward harness explores wider ranges in research mode; production does not retune itself between runs.

Maximizer breakout book (added on top). Maximizer runs a second, separate book: roughly fifteen full-notional breakout slots whose entries fire only in a rotating-bull regime, each held on a fixed ~29-trading-day time-stop rather than the trailing stop, with total exposure scaled continuously by a volatility target. Preserver does not run this book (Section 09).

So the two tiers share the base entry set and diverge in exactly two places: the intensity of the capitulation overlay (Section 08), and whether the breakout book is running — Maximizer only (Section 09).

> **Research vs. production.** The walk-forward harness can re-optimize parameters on a rolling window using Bayesian methods — research capability that informs which parameter set gets locked. Production then runs the locked parameters until a new validation pass justifies replacement. The system does not silently retune itself in front of subscribers.

Three-factor convergence



All three sets must overlap for an entry to fire.

Market regime *intelligence.*

The regime classifier reads SPY price action, VIX levels, market breadth, and new-highs data, and assigns the current environment to one of seven distinct states. The output drives the cash/invested decision and feeds the capitulation overlay.

Seven regimes.

Strong Bull
SPY clearly above 200-day; VIX low; broad participation.

Weak Bull
SPY above 200-day but only modestly; volatility moderate.

Rotating Bull
SPY choppy with positive 1-year trend; sector rotation in play.

Range-Bound
SPY within a few percent of its 200-day; flat trailing return.

Weak Bear
SPY below 200-day, elevated VIX, deteriorating breadth.

Panic / Crash
SPY meaningfully below 200-day; VIX in the upper quantile.

Recovery
Bouncing from lows; SPY above 50-day but still recovering.

Forecast
A 1–2 week transition outlook layered on the static state.

Scoring weights.

Regime detection uses weighted scoring across condition indicators. Higher weights mean that indicator carries more influence:

INDICATOR	WEIGHT	ROLE
Long-term trend (252d SPY return)	2.5x	Differentiator: rotating-bull vs. range-bound
SPY vs. 200MA	1.5x	Primary bull/bear boundary
New highs (%)	1.5x	Breadth confirmation of trend
Breadth (% above 50MA)	1.2x	Width of market participation
VIX percentile, trend strength, SPY vs. 50MA	1.0x	Standard conditions

Regime response.

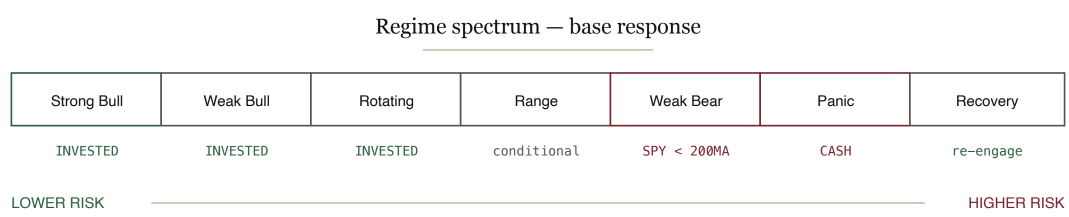
The base regime response is deliberately binary: when the regime engine reads SPY below its 200-day moving average, the system moves to cash; otherwise it stays invested at the locked production parameter set. Layered on top of this is the **capitulation overlay** (Section 08) — a graded cash-raise that keys off a capitulation signal `E` rather than the binary boundary, and whose intensity is one of the two things that separate Preserver from Maximizer. Cascade Guard runs alongside (Section 10).

The regime read also gates the other tier differentiator. In **rotating-bull** specifically, Maximizer's breakout book (Section 09) switches on and enters fresh breakout names; in every other regime it proposes nothing new and its held names age out. So the same regime engine that raises cash in a panic is what turns Maximizer's offense on when the environment rewards it.

REGIME	BASE ACTION	NOTES
Strong Bull	Invested	Locked parameters
Weak Bull	Invested	Locked parameters
Rotating Bull	Invested	Locked parameters · Maximizer breakout book fires here
Range-Bound	Invested unless SPY < 200MA	Locked parameters
Weak Bear	Cash if SPY < 200MA	Capital preservation
Panic / Crash	Cash	Overlay raises cash hardest here

REGIME	BASE ACTION	NOTES
Recovery	Invested	Re-engagement on 50MA reclaim

Walk-forward optimizer choices in research mode are sometimes shown per regime. The live base response is binary cash-or-invested; the capitulation overlay then modulates exposure continuously. Together with the Maximizer breakout book — gated to rotating-bull — it is what differentiates the two tiers.



A binary cash/invested base, anchored on SPY vs. 200-day — the overlay then grades exposure.

Transition forecasting.

Beyond static classification, the regime engine computes indicator trajectories to predict where the market is heading in the next one-to-two weeks:

- **VIX delta (5-day).** Rising VIX boosts bearish regime probabilities.
- **Trend acceleration.** Recent 5-day return compared to the prior 5-day return.
- **200MA proximity.** Within 2% of the 200-day raises transition uncertainty.
- **Sharp moves.** 5-day returns exceeding plus-or-minus 5% boost crash and recovery probabilities.

The forecast collapses to a recommended posture: `stay_invested` (SPY > 200MA) or `go_to_cash` (SPY < 200MA), with the overlay modulating between them.

Trailing stops & *exits.*

The trailing stop is calculated from the high water mark — the highest price since entry, not the entry price itself. Stops ratchet up as a position appreciates and never walk back down. Both tiers use the same 30% trailing stop on the base book; Maximizer's breakout positions instead exit on a fixed hold rather than a trailing stop (Section 09).

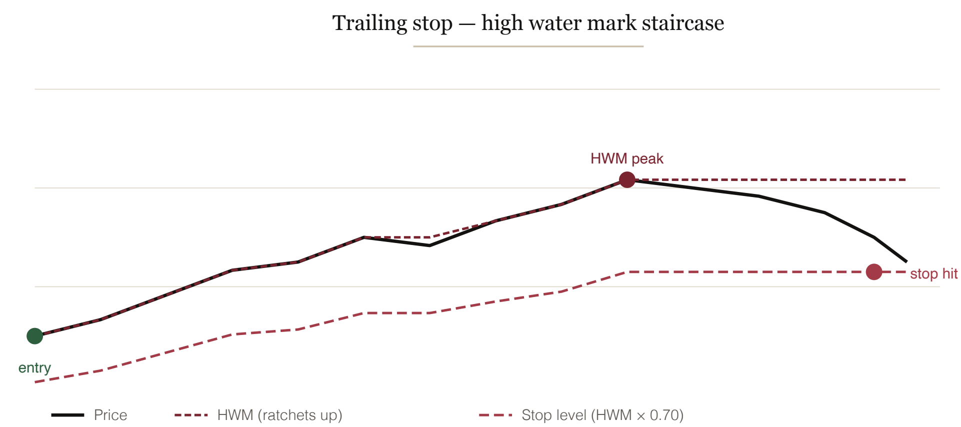
```
# High water mark and trailing stop level
HWM        = max(entry_price, stored_highest, current_price)
stop_level = HWM × (1 − 0.30)

# Exit and warning conditions
SELL    when price ≤ stop_level
WARN    when (price − stop_level) / price < 0.03
```

Exit priority.

1. Regime / overlay exit — when the capitulation overlay raises cash or the regime read flips to `go_to_cash`, positions are trimmed or closed regardless of P&L; capital preservation overrides everything else.
2. Trailing-stop hit — price at or below HWM × 0.70; closes the position.
3. Warning zone — within 3% of the stop level; triggers a subscriber alert, no automatic action.
4. Regime warning — SPY approaching its 200-day average; advisory only.
5. Rebalance rotation — at the periodic universe re-ranking, an existing position has dropped out of the top tier and a stronger candidate displaces it. The boundary itself is not an exit trigger; the rank change is.

This cascade governs the base book both tiers hold. Maximizer's breakout positions follow a separate rule — a fixed ~29-trading-day hold-to-exit rather than a trailing stop (Section 09) — while still subject to the regime and Cascade Guard pauses above.



Stops follow the price up — never down.

The capitulation *overlay.*

This is the first of the two mechanics that separate the tiers — the Maximizer breakout book (Section 09) is the other. Both tiers hold the same base momentum book; the overlay raises cash during capitulation — harder for Preserver, lighter for Maximizer.

What it does.

The overlay watches for capitulation conditions — the environment where the momentum trade tends to unwind violently and where naive trend-following gets whipsawed. When a capitulation signal `E` crosses its threshold, the overlay raises cash on the existing book rather than force-rotating positions. It is a **graded, partial** cash-raise, not a hard on/off switch: over a typical year it fires as a handful of partial de-risks, then re-deploys as conditions normalize.

We chose the overlay over the alternatives deliberately. A hard-rotate rule recovered the same protection but demanded roughly eight full portfolio rotations a year — tax-brutal and execution-heavy in a taxable account. A confirmation-filter approach proved fragile. The overlay keeps the momentum book intact, raises cash only in capitulation, and stays tax-light and deliverable to subscribers who execute at their own broker.

> **The overlay difference in one line.** Preserver runs the overlay at a lower capitulation threshold and a deeper cash-raise; Maximizer runs it at a higher threshold and a shallower raise — same base book, different exposure through a drawdown. (Maximizer's other differentiator, the breakout book, is Section 09.)

Warm-start requirement.

The overlay reads the book's recent equity history to size the cash-raise. That makes warm-start non-negotiable in production: a cold launch with zero equity history leaves the brake blind for weeks, which historically turned a bad-timed launch into a materially worse drawdown. Production seeds the equity history from the validated walk-forward equity curve so the overlay is live from day one. This applies to both tiers, and especially to Maximizer, whose higher beta punishes a blind window hardest.

Down-capture — what the overlay buys.

METRIC	PRESERVER	MAXIMIZER	SPY
2008 full year	+0.1%	+0.1%	−37.7%
2020 full year	+9.2%	+34.9%	+15.2%
2022 full year	−11.2%	−6.4%	−19.9%
Down-month capture	−1.05	−0.97	−3.85
Longest underwater	2.2y	3.4y	5.4y

Overlay walk-forward, 2007–2026 (vintage 2026-07-31). The overlay is why both tiers sidestep the worst of a crash while participating in the recovery.

The overlay is the reason 2008 comes in essentially flat for both tiers while the index loses more than a third, and the reason the Preserver drawdown lands near –12.9% over five years against SPY's –25.4%. It is the mechanical form of the product thesis: momentum, built around the drawdown.

The Maximizer breakout *book.*

The capitulation overlay explains how the two tiers behave through a drawdown. It does not, by itself, explain why Maximizer targets roughly 31% a year against Preserver's 13%. That gap comes from a second, separate engine only Maximizer runs: a gated breakout book, layered on top of the Preserver base.

> **Maximizer is two delineated layers, not one book turned up.** A Maximizer subscriber receives the full Preserver signal set *plus* a standing breakout book. The two are kept separate on purpose — the breakout layer has its own entries, its own holding rule, and its own exposure control, and it carries no base-momentum leg of its own, so it never double-counts the Preserver names a subscriber already holds.

Gated entries — it only hunts when the regime pays.

The breakout book enters only on days the regime engine (Section 06) routes to breakout — the rotating-bull environment where fresh 50-day-high breakouts tend to follow through. In every other regime it proposes nothing new and any names it already holds simply age out. That regime routing *is* the gate: there is no separate on/off switch, and no way for the book to chase breakouts into a market that isn't rewarding them.

Hold-to-exit, not trail.

Base positions ride a 30% trailing stop and can be held for months. Breakout positions are different: each is entered full-notional into one of a fixed roster of roughly fifteen slots and **held to a time-stop of about 29 trading days**, then rotated out regardless of where it sits. A breakout is a bet on a specific post-breakout window; when the window closes, the position closes. That discipline is what stops the book from quietly turning into a bag of stale names.

A continuous volatility target — the seatbelt.

The book does not run at a fixed size. Its reported exposure is scaled every day by a Barroso-style volatility target: the sleeve's return is multiplied by (target volatility ÷ the book's own trailing realized volatility), lagged so the scale is always computed from history the book has already lived, and capped at full exposure. When breakouts turn violent — exactly when momentum crashes happen — the target automatically trims participation. This is the "offense with a seatbelt" behind the tier's higher-return, tamed-tail profile.

> **Why the split matters for the numbers.** The extra return Maximizer shows in strong-trend years — its +34.9% in 2020 against Preserver's +9.2%, for instance — is largely the breakout book working in rotating-bull, not merely a lighter overlay setting. The construction is validated the same way everything else here is: it was reconciled penny-to-penny against the walk-forward reference before it went to production.

Warm-start applies here too.

Like the overlay, the volatility target reads the book's own equity history to size exposure, so it must be warm-started in production from the validated walk-forward curve. A cold launch would leave the target blind precisely when Maximizer's higher beta needs it most.

Cascade Guard — the portfolio *circuit breaker.*

Most position-level systems can be undone by a single bad day — multiple stops fire, new positions enter into the same fall, and the cycle repeats. Cascade Guard exists to interrupt that pattern.

What it does.

When several positions hit their trailing stop on the same day, Cascade Guard pauses all new entries for ten trading days. The threshold and pause length are conservative, and the mechanism is deliberately rare — it fires on cascade events, not on ordinary drawdowns. It runs on both tiers.

> **Live in production.** Cascade Guard has been a fixture of the walk-forward harness for some time; it runs in production with state persisted to S3 (`s3://rigacap-prod-price-data-149218244179/cb-state/<portfolio_type>.json`). Moving it live closed one of the walk-forward-to-production parity gaps.

Mechanics.

LEVER	SETTING	BEHAVIOR
Trigger threshold	3 same-day trailing-stop hits	Cascade detected

LEVER	SETTING	BEHAVIOR
Pause length	10 trading days	No new entries during pause
Existing positions	Unaffected	Continue to be monitored normally
HWM & regime checks	Continue to run	Stops still fire if hit
Re-engagement	Automatic	Pause expires; ensemble resumes

What it adds.

Cascade Guard is validated by counterfactual: the walk-forward run is repeated with the safeguard disabled and the results compared. The drawdown impact is essentially neutral, so we frame Cascade Guard as a re-entry discipline, not as drawdown protection — the capitulation overlay and regime exit are what handle drawdown. The value is asymmetry: the cost during quiet periods is zero, and the benefit during a cascade is avoiding the second wave of re-entries into the same falling market.

Closes a parity gap

Cascade Guard had been a harness-only mechanism. Production now runs the same logic so subscribers experience the lever the walk-forward measured.

Conservative by design

The trigger and pause length are deliberately not aggressive. The mechanism's job is to interrupt cascades, not to frequently override the strategy.

Intraday *monitoring.*

Once positions are open, the system checks live prices every five minutes during market hours. The job is HWM tracking and regime-change detection — not minute-by-minute trading.

Schedule.

PARAMETER	VALUE
Hours	`Monday–Friday, 09:00–15:55 ET`
Interval	`Every 5 minutes`
Price source	`Alpaca latest quote (SIP), yfinance fallback`
Alert deduplication	`(position_id, action, date)`

Trailing-stop exits run end-of-day.

After research showed intraday cadence cost roughly 17 percentage points annualised versus end-of-day evaluation, trailing-stop exits are evaluated end-of-day. HWM tracking and regime-exit checks continue to run every five minutes during market hours; the highest price seen each session is captured for the next day's stop calculation, and a regime transition can still close the book intraday. The 5-minute monitor is a state-tracking surface, not an execution loop. The behavior is controlled by a single environment flag (`INTRADAY_TRAILING_STOP_ENABLED=false` in production).

> **Why end-of-day stops.** Intraday execution is exposed to data anomalies and brief flash moves that resolve before the close. Evaluating the trailing stop on the daily close removes that noise without sacrificing the discipline — the stop still fires the same day, just on a confirmed price.

Watchlist intraday signal detection.

The monitor also watches the dashboard’s watchlist — candidates approaching the timing threshold but not yet through it. If a watchlist candidate clears the threshold intraday, the system promotes it to a fresh signal, removes it from the watchlist, alerts subscribers who do not already hold it, and updates the S3 dashboard JSON in place.

Position *management.*

The model portfolio service maintains parallel ledgers — live Preserver and Maximizer books that mirror production behavior, a walk-forward book for performance measurement, and a signal track-record book that logs every signal the system issues.

Production parameters (shared base).

20	4.5%	30%	Overlay
MAX POSITIONS	POSITION SIZE (INVERSE-VOL)	TRAILING STOP	TIER DIFFERENTIATOR

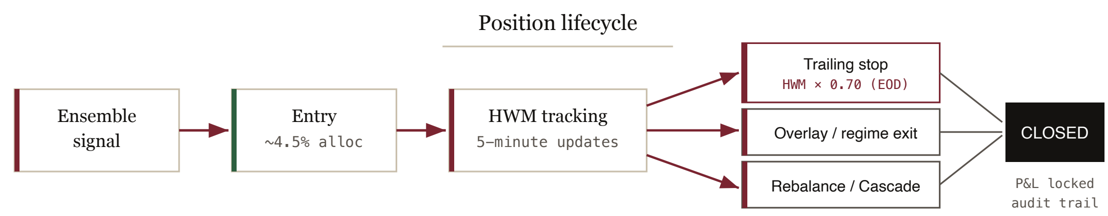
Entry logic.

- Read fresh ensemble signals from the dashboard cache (S3 JSON).
- Enter currently-qualifying candidates; skip symbols already held.
- Sort by ensemble score, highest first.
- Enter up to `MAX_POSITIONS − open_count` new positions, subject to the tier's overlay exposure ceiling.
- Allocate roughly 4.5% per position, weighted by inverse volatility so calmer names carry more.
- Minimum allocation $100 to avoid sub-share rounding.

Exit paths.

EXIT REASON	TRIGGER	BOOK
`trailing_stop`	Daily close ≤ HWM × 0.70	Live + WF
`hold_exit`	Fixed ~29 trading-day time-stop on a breakout position	Live + WF (Maximizer breakout book)
`overlay_cash_raise`	Capitulation signal E over threshold	Live + WF (tier-scaled)
`regime_exit`	`go_to_cash` recommendation	Live
`rebalance_exit`	Period boundary (walk-forward harness)	Walk-forward
`cascade_pause`	Cascade Guard active during entry window	Live + WF

Re-ranking boundaries. "Rebalance" here means a universe re-ranking, not a portfolio shuffle for the subscriber. The system periodically re-runs the entry gauntlet across the full universe, identifies the current top-tier candidates, and checks whether existing held positions still qualify. Positions that still rank carry forward; ones that have dropped out rotate out. Carry-on semantics outperform force-close, so the system carries qualifying positions across windows rather than mechanically liquidating at the boundary.



A position is a state machine with three exit families.

Walk-forward to production — exact *parity.*

The walk-forward result is only realizable if production runs the same strategy the walk-forward validated. Every mechanic — ranking universe, HWM tracking, stop trigger, position sizing, capitulation overlay, and the Maximizer breakout book (gated entries, hold-to-exit, vol-target) — is audited for parity, and any divergence is treated as a defect that costs subscribers real returns across both tiers.

Most of these alignments were in place at launch. A handful drifted as features were added incrementally and were realigned after audit. The table below is the post-audit state.

Parity table.

MECHANIC	WALK-FORWARD DEFAULT	PRODUCTION LIVE
Ranking universe	top-N by 60-day avg volume, recomputed per period	matched (a prior N=100 drift to 200 was corrected)
HWM source	day's close	day's close (from price cache)
HWM persistence	in-process across periods	database commit on every pass (unconditional)
Stop trigger	close ≤ HWM × (1 – stop %)	close ≤ stop level (intraday firing intentionally disabled)
Trailing stop %	30% (both tiers)	30% (both tiers)

MECHANIC	WALK-FORWARD DEFAULT	PRODUCTION LIVE
Max concurrent positions	20	20
Position sizing	~4.5% per slot, inverse-volatility weighted	~4.5% per slot, inverse-volatility weighted
Entry gauntlet	timing + momentum + confirmation + anti-squeeze	identical filters
Capitulation overlay	graded cash-raise on signal E, tier-scaled	identical, warm-started from WF equity curve
Carry positions across rebalance	true (positions survive period boundaries)	true (live book is always-on)
Cascade Guard	3 same-day stops → 10-day pause	identical, state persisted to S3
Breakout book — entries (Maximizer)	gated to rotating-bull; ranked breakout candidates only	identical gating
Breakout book — hold-to-exit (Maximizer)	fixed ~29 trading-day time-stop	identical
Breakout book — vol-target (Maximizer)	Barroso target ÷ trailing realized vol, lagged, capped 1.0	identical, warm-started from WF equity curve

The audit method.

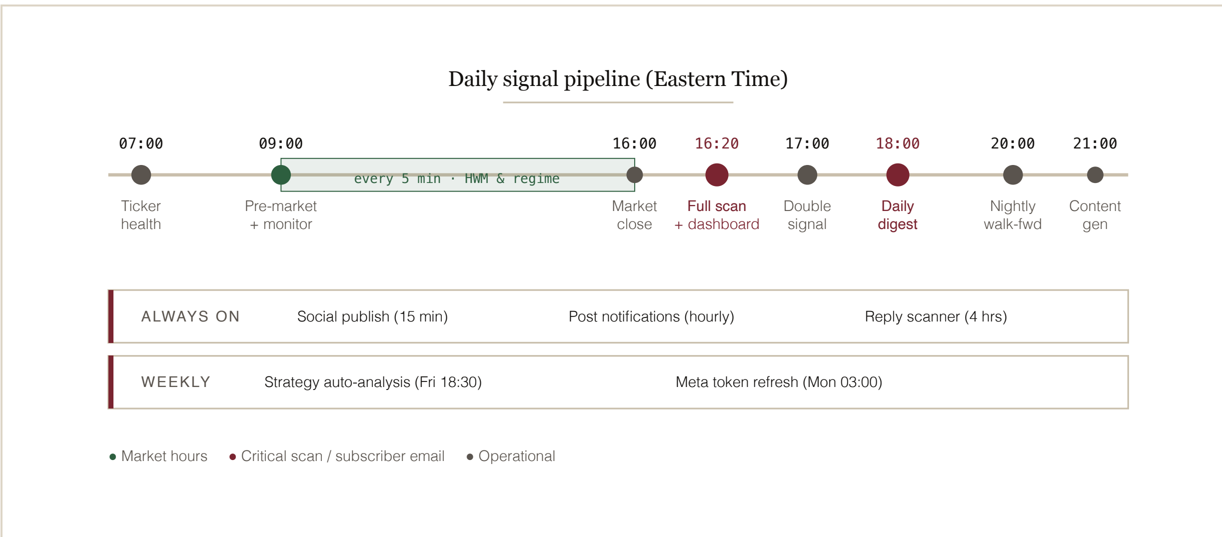
Parity is not assumed; it is measured. The largest single realization gap discovered in audit was the ranking universe: production had filtered to a smaller most-liquid set than the walk-forward. A liquidity-rank audit of a representative run found that a large majority of simulated trades — including several of the strongest winners — came from symbols ranked below the production cutoff. The fix was a one-line configuration change; the discovery required reading the trade list against the same ranking logic and counting where each entry came from.

Because both tiers share the engine, a parity defect is doubly expensive — it corrupts the marketed edge for Preserver *and* Maximizer at once. This kind of mechanical audit runs whenever a new lever is added to either side — harness or production — before the lever is allowed to influence published results.

> *Code parity is not runtime parity. Diff production's actual config — never assume it.*

The daily *timeline.*

A coordinated set of scheduled jobs runs each trading day, from morning health check through nightly content generation. Every job is managed by EventBridge with US Eastern timezone awareness.



Alert types.

ALERT	TRIGGER	AUDIENCE	CHANNEL
Daily digest	18:00 ET, trading days	All subscribers	Email + push
Sell alert	Trailing stop, overlay, or regime exit	Position holders	Email + push
Warning alert	Within 3% of stop level	Position holders	Email + push

ALERT	TRIGGER	AUDIENCE	CHANNEL
Double signal	Timing + momentum cross within 3 days	All subscribers	Email
Intraday signal	Watchlist candidate clears threshold	Subscribers (not holding)	Email
Ticker health	Data issues detected at 07:00	Admin	Email

Email deliverability: SPF, DKIM, and DMARC authentication, RFC 8058 List-Unsubscribe headers for one-click unsubscription, and per-category preference controls.

Walk-forward *methodology.*

Walk-forward testing is the standard for trading-system validation. Rather than optimizing across all available data and baking in hindsight, it locks parameters in advance and tests forward into data the optimizer has never seen.

Methodology.

- Window. A single continuous 21-year path (2007–2026), price returns, plus a rolling 5-year read for the current headline figures.
- Optimization. Parameters are locked, then applied forward; no per-period retuning.
- Warm start. One canonical full-history equity curve; windows are sliced from it so the capitulation overlay always sees a real equity history (never cold).
- Trading. Enters fresh ensemble signals as they fire; the universe is periodically re-ranked and existing positions are checked against the refreshed top tier.
- Position carry. Carry-on semantics: qualifying positions survive period boundaries rather than being mechanically liquidated.
- Both tiers. Preserver and Maximizer are run through the same day-step harness; only the overlay intensity differs.
- Data integrity. Survivorship-free, point-in-time data from 2016 onward; the pre-2016 extension carries a survivorship caveat.

Research maps to production, penny-to-penny. Every research result is re-validated through the production day-step on the same warm equity curve before it is trusted or marketed. Only exposure-scaling constructions (raising cash) reproduce in the single-pool book; idealized sleeve-capture return-streams do not, and are rejected.

Zero hindsight bias. At each step the harness knows nothing about what happens next. Parameters are fixed in advance; the strategy then trades forward into unseen market conditions. The published figures are out-of-sample by construction.

What the walk-forward *shows.*

Two tiers off one engine, over a full cycle. The crisis years and the bad years are on the record alongside the averages — that is the point of the product.

Headline numbers.

SERIES	CAGR	SHARPE	WORST DRAWDOWN
Preserver — 5yr	13.0%	1.28	-12.9%
Maximizer — 5yr	31.4%	1.51	-14.9%
SPY — 5yr	*14.2%*	—	*-25.4%*
Preserver — 21yr	7.7%	0.87	-13.7%
Maximizer — 21yr	13.5%	0.93	-20.8%
SPY — 21yr	*9.8%*	—	*-55%*

Source: continuous walk-forward path, price returns (vintage 2026-07-31). Data from 2016 onward is survivorship-free and point-in-time; the pre-2016 extension carries a survivorship caveat. The live model book began June 2026 — all figures here are walk-forward.

Typical windows.

Preserver

Typical 12-month +10.7% (77% of windows positive); typical 24-month +20.8%. Longest underwater stretch 2.2 years against SPY’s 5.4. Beats SPY in 36% / 17% / 14% of rolling 1/3/5-year windows.

Maximizer

Typical 12-month +26.5% (77% positive); typical 24-month +57.5%. Longest underwater stretch 3.4 years. Beats SPY in 50% / 45% / 31% of rolling 1/3/5-year windows.

Where the edge actually is.

The edge is risk-adjusted, not raw. Over the full cycle both tiers cut the worst drawdown to roughly a quarter of the index’s — near –13% for Preserver and –21% for Maximizer against SPY’s –55% — while down-month capture runs –1.05 (Preserver) and –0.97 (Maximizer) versus –3.85 for the index. The system is built to be held through a full cycle, including by investors who would panic-sell anything that loses half its value. That is the thesis: momentum, built around the drawdown.

Crisis capital preservation

2008: both tiers +0.1% while SPY fell –37.7% (overlay in cash). 2022: Preserver –11.2% and Maximizer –6.4% against –19.9%. The clearest empirical evidence the overlay earns its keep.

Bull participation

2020: Preserver +9.2%, Maximizer +34.9% against +15.2% for SPY — back in for the recovery after sidestepping the worst of the crash.

How we trade.

On the base book, most positions are exited at the trailing stop, by design. The system does not need to be right most of the time; it needs winners to compound and losers to be cut on a fixed rule — what the wide 30% trailing stop, the capitulation overlay, and the exit priority deliver across up to twenty inverse-volatility-sized positions. Maximizer layers the breakout book on top, where positions instead run to a fixed

~29-trading-day hold with exposure governed by the vol-target — a different exit discipline for a different job.

Winners run, losers cut

The wide trailing stop lets profitable positions compound through normal volatility rather than capping them, while exits on the rule keep losses bounded.

Diversification, not concentration

Twenty smaller positions instead of a concentrated handful. The bar at entry stays high — the gauntlet rejects far more candidates than it admits — but no single name can sink the book.

Source data: continuous walk-forward path (2007–2026, price returns) on clean indicator data; headline 5-year and 21-year figures vintage 2026-07-31. Data from 2016 onward is survivorship-free and point-in-time; pre-2016 data carries a survivorship caveat. The live model book began June 2026. Production parameters (shared base): up to twenty positions sized at roughly 4.5% each with inverse-volatility weighting, a 30% trailing stop, the market-regime filter, Cascade Guard, and the timing-and-momentum ensemble described in this document. The Preserver and Maximizer tiers differ in the capitulation-overlay intensity and in the Maximizer breakout book (Section 09).